Grade 10 Cycle TEST
September 2020
MEMO

Question One

1.1. B
1.2. C
1.3. A
1.4. ~~D~~ B
1.5. D

(2 x 5 = 10)

Question 2

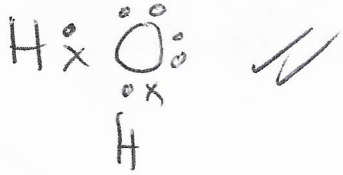
2.1.1. E ✓ highest m.pt + b.pt + poor electrical conductivity. ✓ (2)

2.1.2. B ✓ Conducts electricity as a solid + a liquid ✓ (2)

2.1.3. C ✓ Conducts elec as a ~~solid~~ liquid but not solid ✓ (2)

2.1.4. ~~A~~ ~~F~~ ✓ Lowest m.pt + b.pt + melts below room temp ✓ (2)

2.2.1.



(2)

2.2.2. Weak force of attraction between molecules, ions + atoms of noble gases (2)

2.2.3. Hydrogen bonding (1)

2.2.4. London forces (1)

2.2.5. H_2 bonding stronger ✓ than London forces ∴ more energy required ✓ to overcome. (2)

2.2.6. CO_2 also has London forces ✓ but these are stronger than those of CH_4 as bigger molecule mass ∴ more electrons ∴ more temporary dipoles ✓ ∴ more energy required to overcome. (3)

2.2.7. (i) CO_2 ✓✓ (2)

(ii) CO_2 requires energy ✓ to change phase from solid to gas ⇒ this is removed ✓ from the H_2O ∴ dec-n temp. (2)

2.2.8. (i) The amount of heat needed to change the state of 1kg of a substance without a its temp. (2)

(ii) It is used to overcome ✓ the IMFs holding H_2O molecules in the solid phase ✓ (2)

(iii) The energy added increases ✓ the av. kinetic energy ✓ of water molecules once all are in the liquid phase (2)

Question 3

3.1. The total internal energy contained in an object ✓✓ (2)

3.2. To determine ✓ wether the efficiency of a heat exchanger depends on the material used to make the pipe. ✓ (2)

3.3. Type ✓ of material ✓ pipe is made of (2)

3.4. Rate of flow of hot water.
Time after which temp measured
Starting temp of water.
Volume / mass of water in container
ANY ONE (1)

3.5. Copper ✓. Biggest difference in temp ✓ (2)

3.6. The amt of energy required to raise the temperature of 1kg of the substance by 1°C.

3.7. $Q = (4200)(\frac{500}{1000})(23-20)$ ✓ ✓

$= 6300\ J$ ✓ (3)